DAY 16 OF 30

Real-Time Intelligence

What "real time" actually means.
And when you truly need it.

Streaming | Events | Sub-second

Keshav Maheshwari
Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

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THE PROBLEM

A nightly report tells you what already went wrong.

1	**A machine overheats**	You find out tomorrow
2	**Orders pile up**	You see it in the weekly review
3	**A payment fails**	The customer tells you first

Batch data answers "what happened". **Some questions need "what is happening".**

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Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

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THE SIMPLE VERSION

A live scoreboard, not tomorrow's paper.

The newspaper gives you the final score the next morning. The scoreboard changes the moment a goal is scored.

Real-Time Intelligence is **the scoreboard for your business data**.

✓ **Batch**	Collect, then process later
✓ **Streaming**	Process as each event arrives
✓ **The gain**	Seconds instead of hours

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THE MAP

The five building blocks

1. **Real-Time hub**: find every stream in the organisation
2. **Eventstream**: bring events in and shape them on the way
3. **Eventhouse**: store and query huge volumes of events
4. **Real-Time Dashboard**: see what is happening now
5. **Activator**: act automatically when a condition is met

One flow

Capture, store, analyse, visualise, act. All inside Fabric, all on OneLake.

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FOR ENGINEERS

How to think about it

Events, not tables

A stream is an endless series of small messages. You query windows of time, not a finished dataset.

Latency is a requirement, not a default

Ask how fresh the answer really needs to be. Seconds, minutes and hours are very different builds.

It can meet batch in OneLake

Event data can be exposed as Delta tables, so streaming and batch data can be joined.

Different capacity behaviour

Streaming runs all day. Budget for steady usage, not occasional peaks.

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THE HONEST PART

Do you need real time?

YES WHEN	PROBABLY NOT WHEN
Someone acts on the data within minutes	Decisions are made daily or weekly
Sensors, clicks or logs arrive nonstop	The source only updates overnight
Delay costs money or safety	Nobody is watching the screen

> **The catch:** real time is expensive to run and easy to over-buy. If nobody acts faster because of it, an hourly batch gives the same outcome for much less.

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FROM REAL WORK

Real time on a real project

A consumer delivery platform that needed live operational numbers for its leadership.

1. **Event Hub and Eventstream** carried the streaming data in
2. **Volumes** reached terabytes per minute
3. **Eventhouse and KQL** delivered sub-second analytics
4. **The first production use case** shipped in under 14 days

At that scale, the architecture **is** the product.

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Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

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SAVE THIS

Real-Time Intelligence in five lines

1. **Batch** tells you what happened
2. **Streaming** tells you what is happening
3. **Eventstream** in, **Eventhouse** to store
4. **Dashboards** to see, **Activator** to act
5. **Use it** where minutes truly matter

Tomorrow: Eventstream.
Catching data as it happens.

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Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

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